

If we have several years of data, we can calculate a time-weighted return for each year individually, as above. If R_i is the time-weighted return for year i , we calculate an annualized time-weighted return as the geometric mean of N annual returns, as follows:

$$R_{TW} = [(1 + R_1) \times (1 + R_2) \times \dots \times (1 + R_N)]^{1/N} - 1. \quad (6)$$

Example 9 illustrates the calculation of the time-weighted rate of return.

EXAMPLE 9

Time-Weighted Rate of Return

Strubeck Corporation sponsors a pension plan for its employees. It manages part of the equity portfolio in-house and delegates management of the balance to Super Trust Company. As chief investment officer of Strubeck, you want to review the performance of the in-house and Super Trust portfolios over the last four quarters. You have arranged for outflows and inflows to the portfolios to be made at the very beginning of the quarter. Exhibit 14 summarizes the inflows and outflows as well as the two portfolios' valuations. In Exhibit 11, the ending value is the portfolio's value just prior to the cash inflow or outflow at the beginning of the quarter. The amount invested is the amount each portfolio manager is responsible for investing.

Exhibit 14: Cash Flows for the In-House Strubeck Account and the Super Trust Account (US dollars)

	Quarter			
	1	2	3	4
Panel A: In-House Account				
Beginning value	4,000,000	6,000,000	5,775,000	6,720,000
Beginning of period inflow (outflow)	1,000,000	(500,000)	225,000	(600,000)
Amount invested	5,000,000	5,500,000	6,000,000	6,120,000
Ending value	6,000,000	5,775,000	6,720,000	5,508,000
Panel B: Super Trust Account				
Beginning value	10,000,000	13,200,000	12,240,000	5,659,200
Beginning of period inflow (outflow)	2,000,000	(1,200,000)	(7,000,000)	(400,000)
Amount invested	12,000,000	12,000,000	5,240,000	5,259,200
Ending value	13,200,000	12,240,000	5,659,200	5,469,568

1. Calculate the time-weighted rate of return for the in-house account.

Solution:

To calculate the time-weighted rate of return for the in-house account, we compute the quarterly holding period returns for the account and link them into an annual return. The in-house account's time-weighted rate of return is 27.01 percent, calculated as follows: